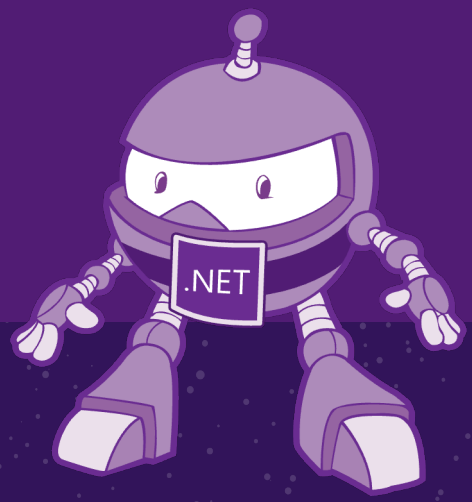
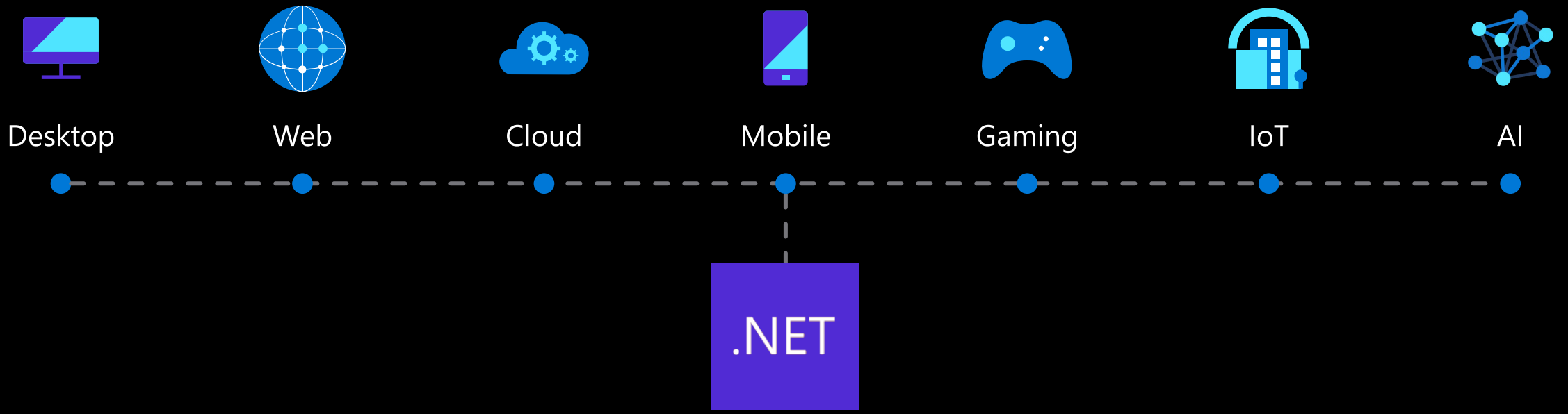
A Roadmap for .NET on Desktop, Mobile, and Web

David Ortinau, Principal Program Manager
@davidortinau
github.com/davidortinau

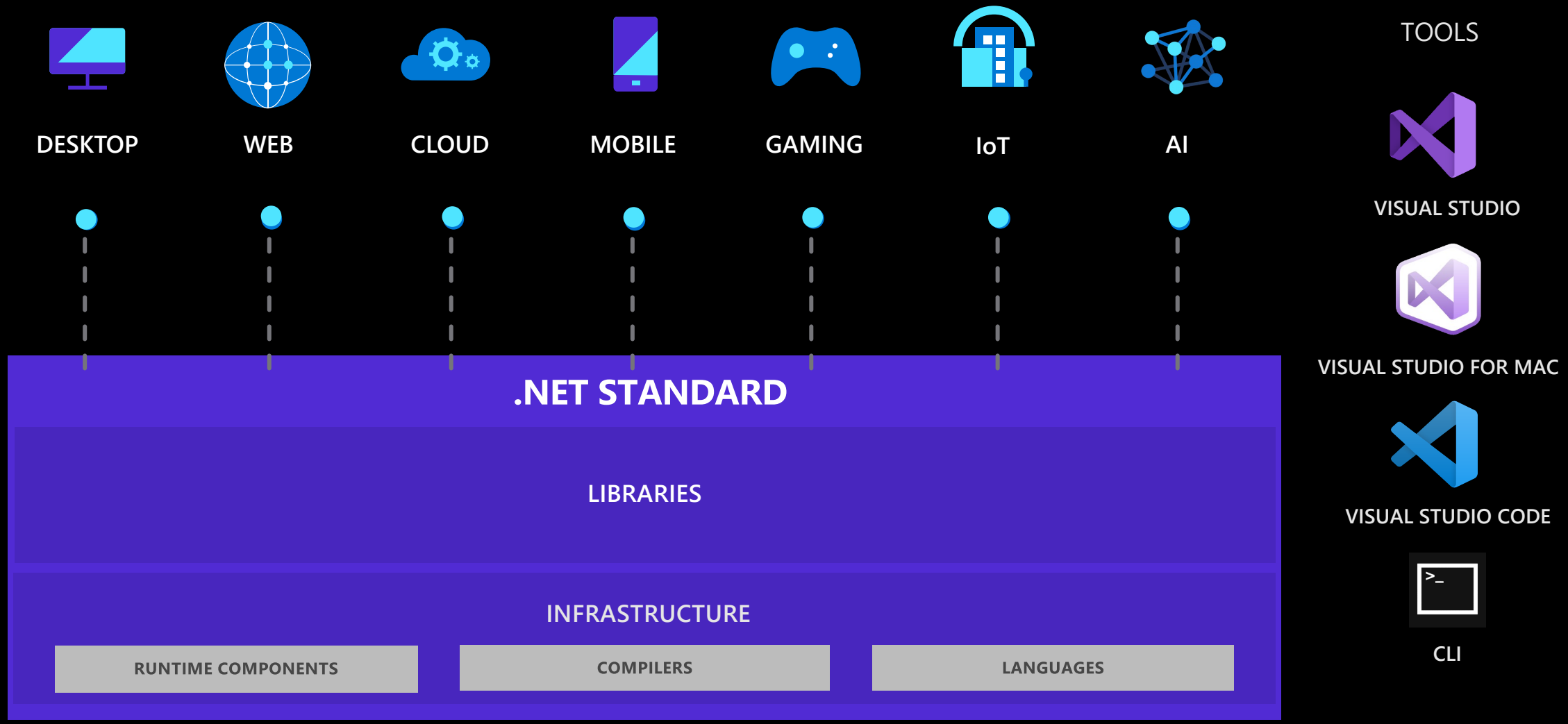


.NET

Your platform for building anything



.NET – A unified platform



.NET ecosystem momentum

+1M
New .NET devs in last year

in Visual Studio

#1 Most Loved
Framework

.NET Core

stackoverflow

Top 30
Highest velocity OSS projects

github.com/dotnet
github.com/aspnet

.NET foundation

Top 5
Language on GitHub

C#



7x
Faster than Node.js

ASP.NET Core

TechEmpower

40%
New to .NET are students

dot.net download survey

.NET open source momentum



87%

contributors outside Microsoft

110,000+

accepted PRs from community

RELEASED

.NET Core 3.1

Long-Term Support (LTS)

WPF and Windows Forms support

Side-by-side support & self-contained EXEs

New C# 8.0 language features

Blazor: Full-stack web development with C# and Razor

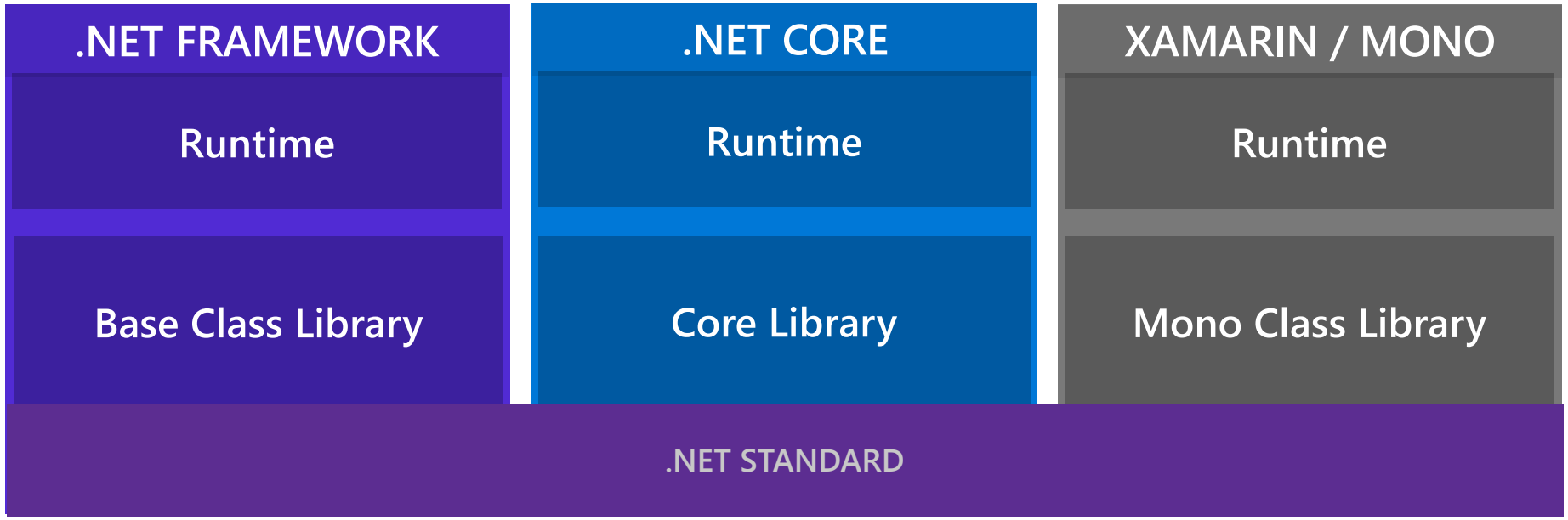


dot.net/get-core3

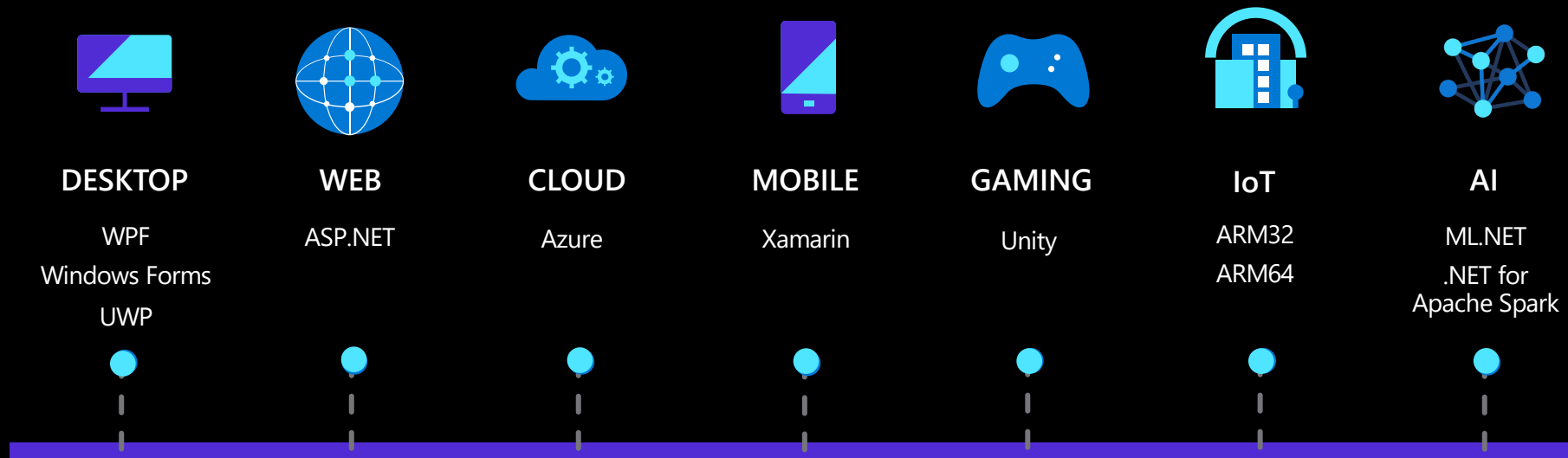
.NET 5



.NET 5


.NET FRAMEWORK
Runtime
Base Class Library
.NET CORE
Runtime
Core Library
XAMARIN / MONO
Runtime
Mono Class Library
.NET STANDARD

.NET – A unified platform



TOOLS

VISUAL STUDIO

VISUAL STUDIO FOR MAC

VISUAL STUDIO CODE

CLI

.NET 5

- *unifies* the .NET Platform across web, mobile, desktop, AI/ML, Big Data, and IoT workloads

- provides a unified set of platform libraries, compilers and tools across workloads

- supports new and modern application models
  - Build cross-platform native apps for any device (with Xamarin.Forms)
  - Build cross-platform web apps (with ASP.NET Blazor)
  - Build high-performance microservices and containerized apps

- simplifies the choices for new developers

.NET 5 Vision – One .NET, A Unified Platform

- Single SDK, One BCL
- Cross Platform Native UI
- Cross Platform Web UI
- Mobile – Xamarin included
- Cloud Native – .NET Orchestration Tools and High-Performance Proxy
- Continual Improvement – Speed, Size, Diagnostics and Azure Services

.NET will attract new and retain current developers because it has the best of breed solutions for all modern workloads

.NET Schedule



- .NET Core 3.0 releases!
- .NET Core 3.1 = Long Term Support (LTS)
- .NET 5.0 release in November 2020
- Major releases every year, LTS for even numbered releases
- Predictable schedule, minor releases if needed

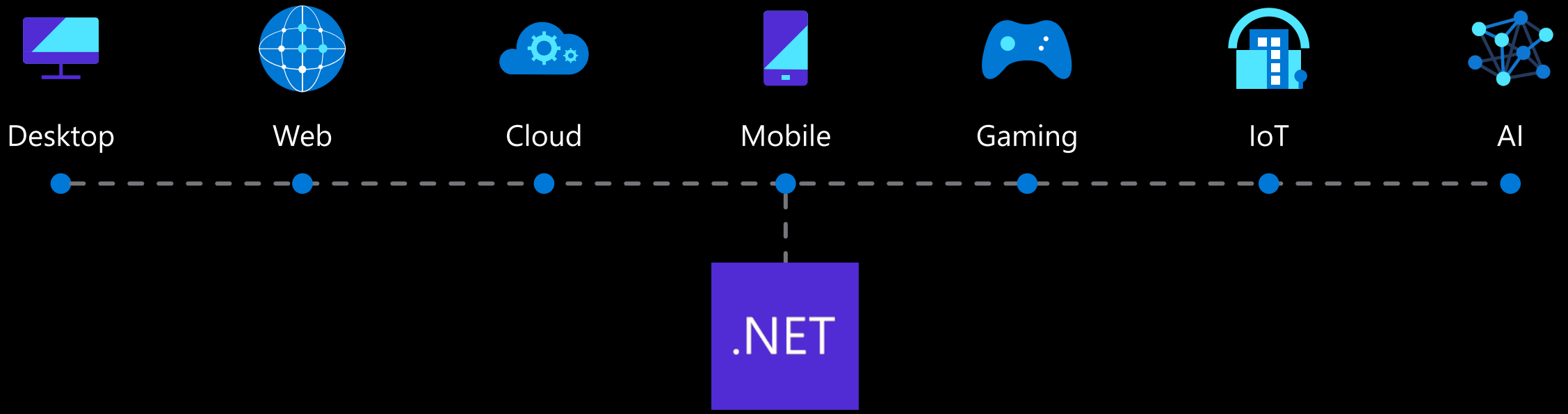
Choose Your Path to Dominance

.NET



.NET

Your platform for building anything



Choosing Your App Model

I like C#

I like HTML

I'm ok with CSS

I know browsers



I like C#

I like XAML

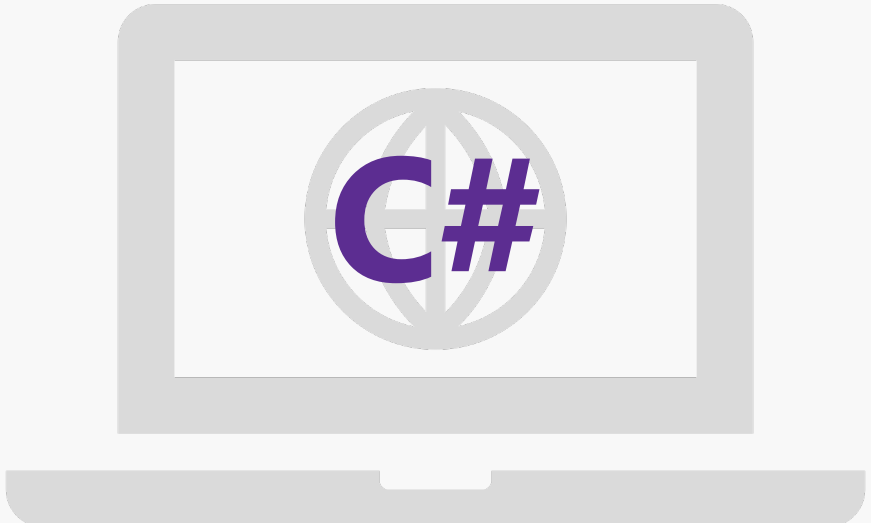
I like Styles

I know devices



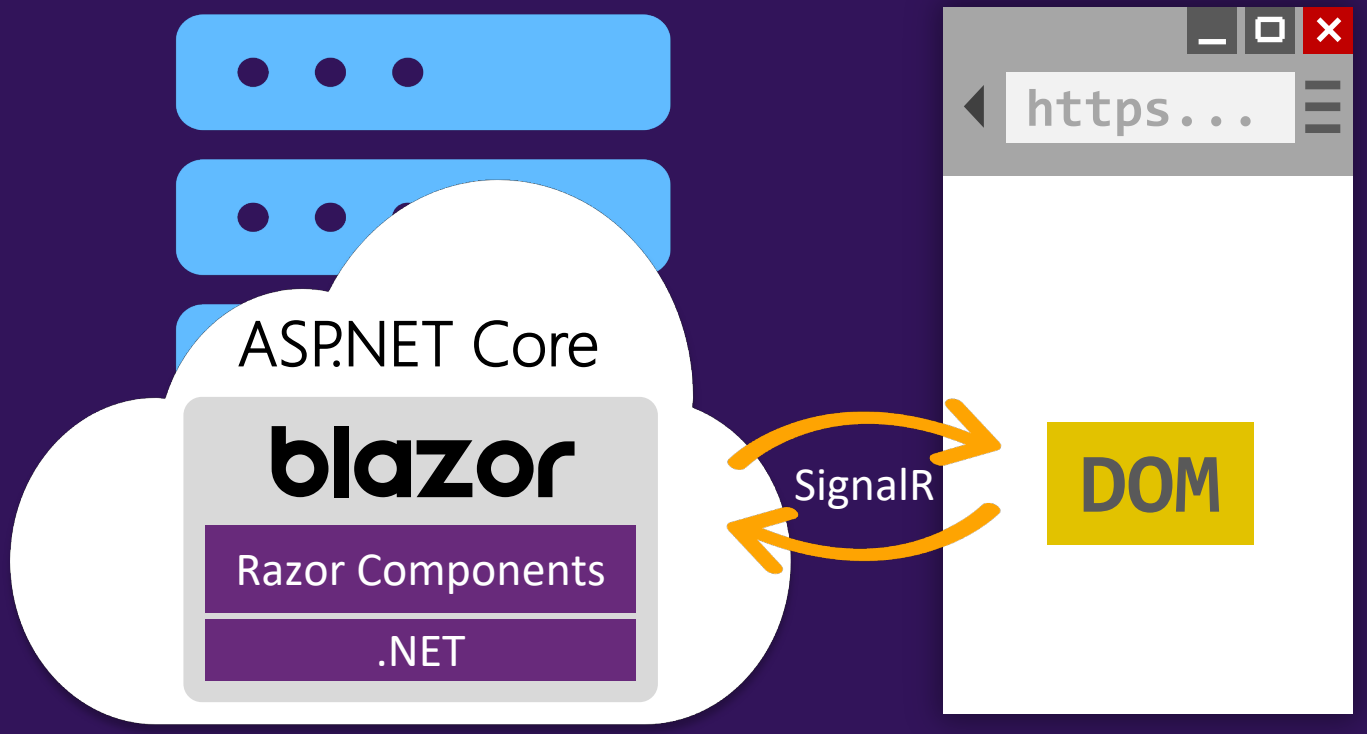
Blazor

- Build client-side web UI with .NET instead of JavaScript
- Write reusable web UI components with C# and Razor
- Share .NET code with both the client and the server
- Call into JavaScript libraries & browser APIs as needed



How Blazor works

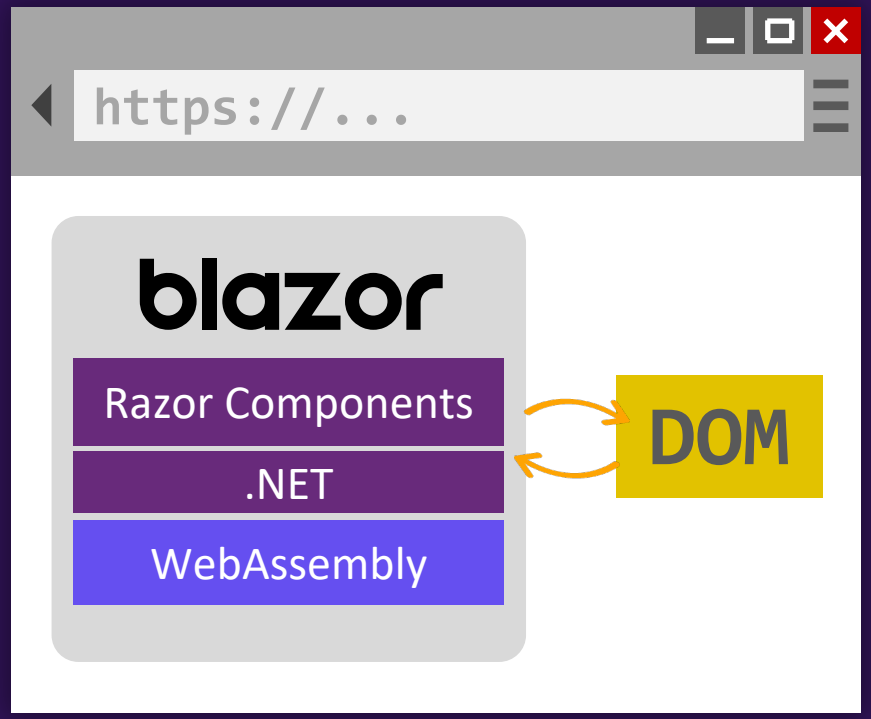
Blazor Server



.NET Core 3.1 LTS

How Blazor works

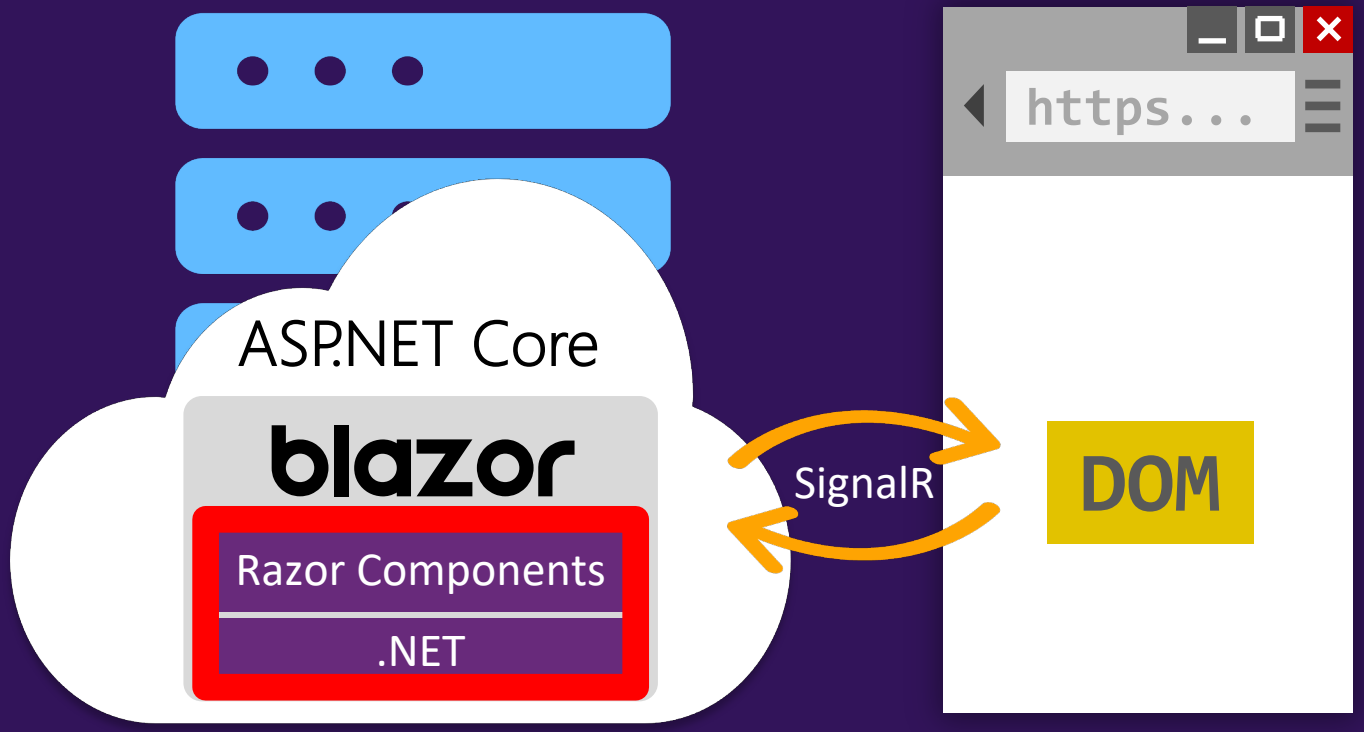
Blazor WebAssembly



Preview

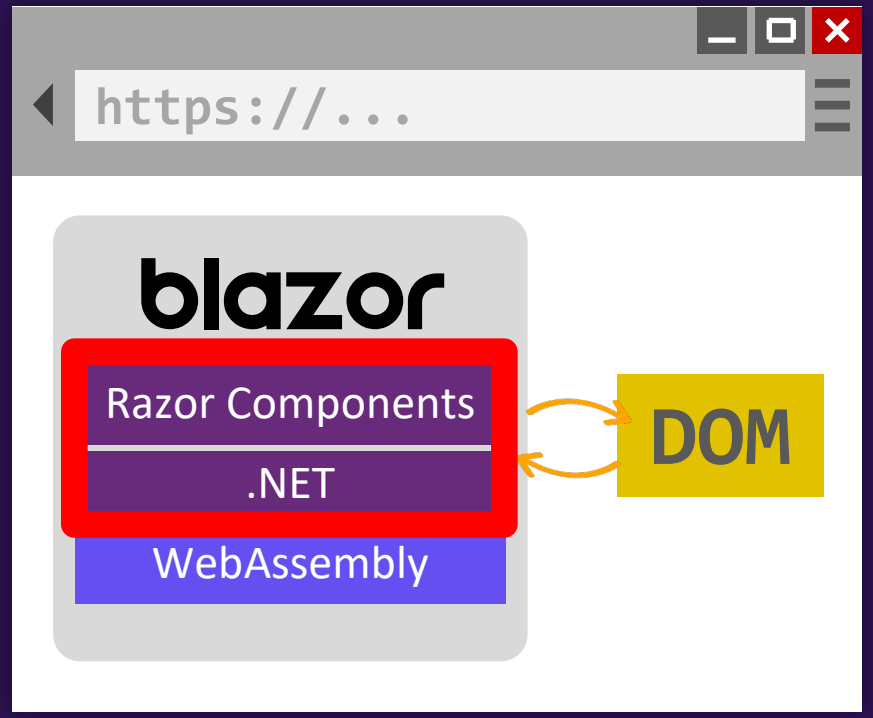
Shared component model

Blazor Server

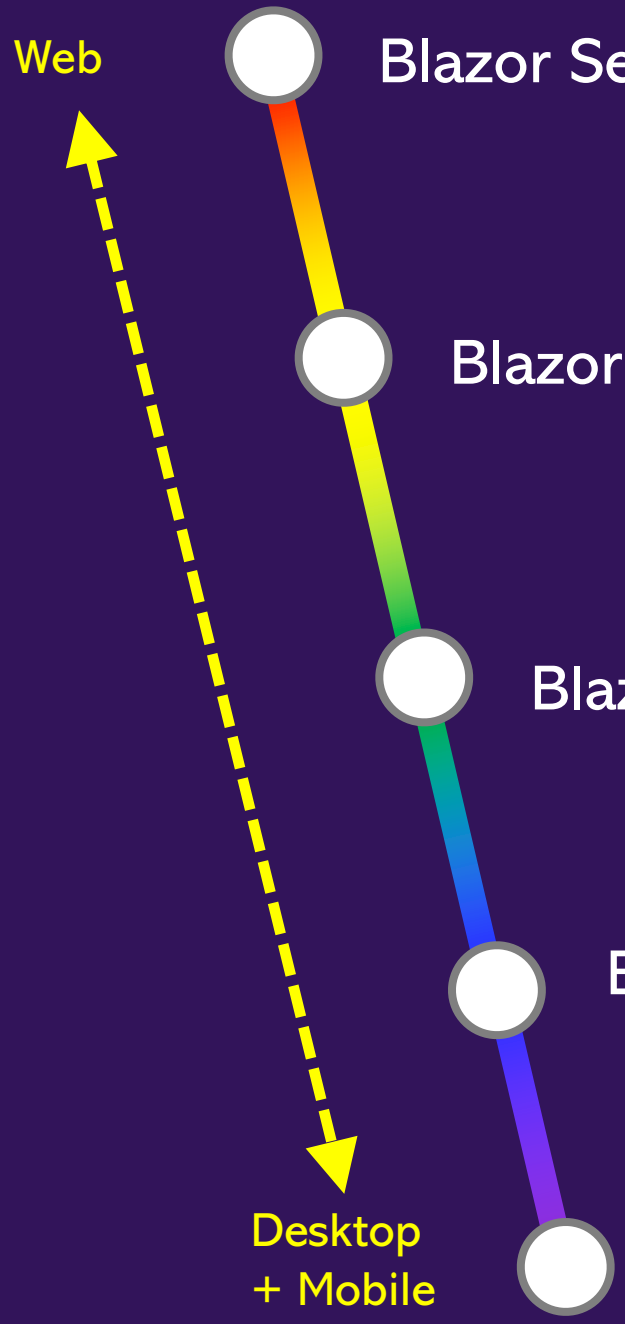


.NET Core 3.1 LTS

Blazor WebAssembly



Preview



Blazor Server

Web app
Every interaction handled on server
Prerendered HTML (optional)

Blazor WebAssembly

Web app with client-side execution
Loaded from web server
Can work offline via Service Worker

Blazor PWA – OS installed

Appears as native app (own window)
Works offline or online

Blazor Hybrid

Native .NET renders to WebView
Appears as native app (own window)
Works offline or online

Blazor Native

Same programming model, but
rendering non-HTML UI

ANNOUNCING

Experimental Mobile Blazor Bindings

Build native mobile apps *with Blazor*

Use your familiar web skills: Razor, C#, CSS

Rich set of native mobile components

100% access to native APIs: GPS, Media, & more

Use existing ecosystem of NuGet libraries for mobile

All in .NET

aka.ms/mobileblazorbindings



Blazor for W

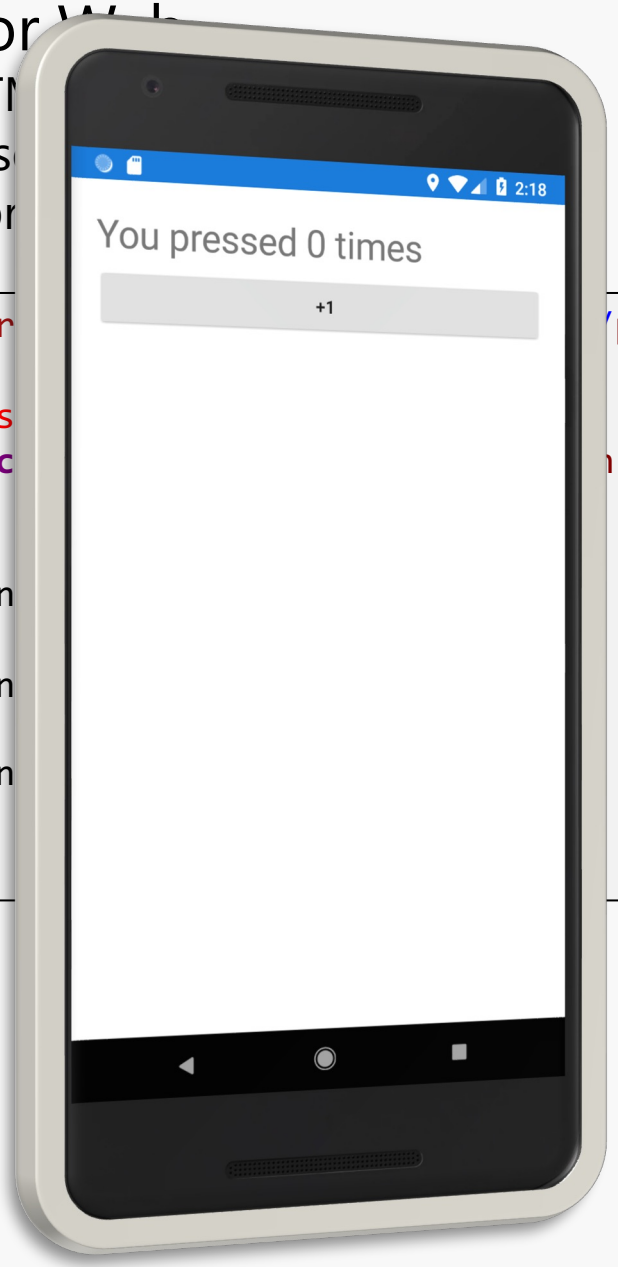
- C# + HTM
- Run on s
- Run in br

```
<p>@("You pr                                    /p>

<button clas
        @onc                                    n>

@code {
    int coun

    void Han
    {
        coun
    }
}
```



Mobile Blazor Bindings

- C# + Mobile UI components
- Run on device natively on Android and iOS

```
<StackLayout>
    <Label FontSize="30"
           Text="@("You pressed " + count + " times")" />
    <Button Text="+1"
            OnClick="@HandleClick" />
</StackLayout>

@code {
    int count;

    void HandleClick()
    {
        count++;
    }
}
```

Get started with Blazor

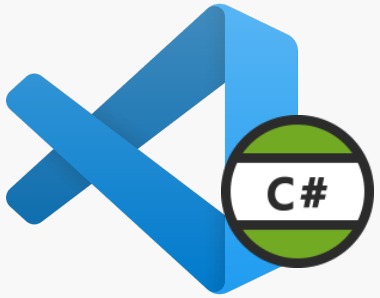
- Go to https://blazor.net
- Install .NET Core 3.1
- Install the Blazor WebAssembly template



Visual Studio
2019 16.4



Visual Studio for Mac
8.4



Visual Studio Code
+ C# extension

https://aka.ms/blazorconf

.NET



With Xamarin, you can build **fast**, **beautiful**, **native** apps in **less time**, with **less code**, within a **thriving ecosystem**

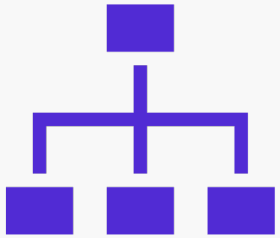
.NET mobile apps with Xamarin



iOS & Android with C#

Native performance, native APIs, native UI

Build native mobile apps with Visual Studio



Cross-Platform

Build cross-platform UI with Xamarin.Forms

Access native APIs from shared code with Xamarin.Essentials

Share code with .NET Standard



Free & Open Source

Xamarin is part of .NET and open source on GitHub



Xamarin

Loved by developers, trusted by enterprise





































































http://xamarin.com/showcase

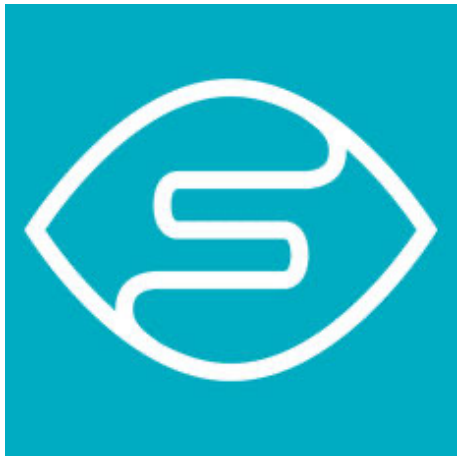


First Party Apps



Microsoft News

Using the Microsoft News app, you'll find best-in-class content from market leaders around the world. Automatically sync your favorite topics across Microsoft News on the web and your mobile apps for quick access to the news you want to read.



Seeing AI

An app that narrates the world around you. Designed for the low vision community, this research project harnesses the power of AI to describe people, text and objects



Azure App

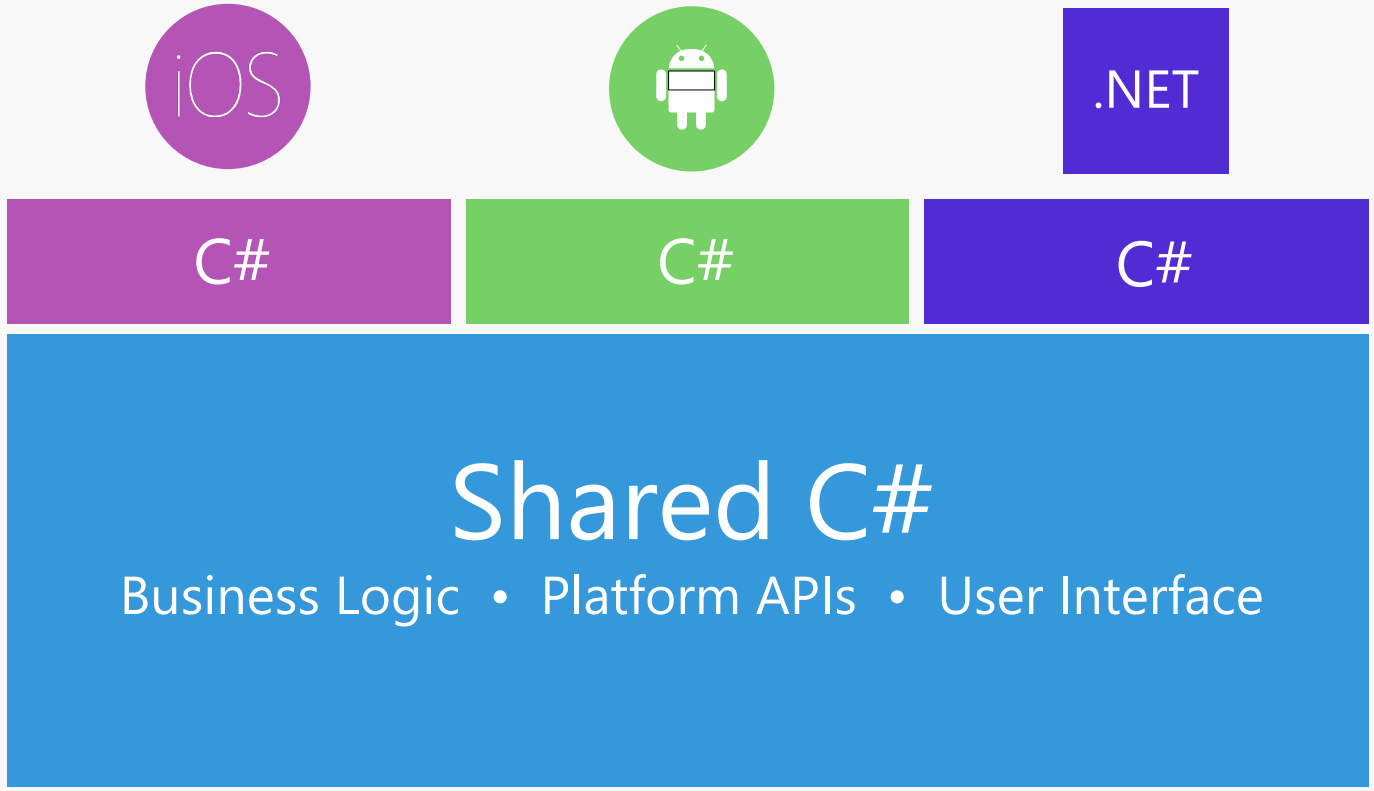
With the Azure mobile app, you don't need to be in front of your computer to keep an eye on your Azure resources such as VMs and web apps. Stay connected no matter where you are from your iOS or Android mobile device.



O365 Admin

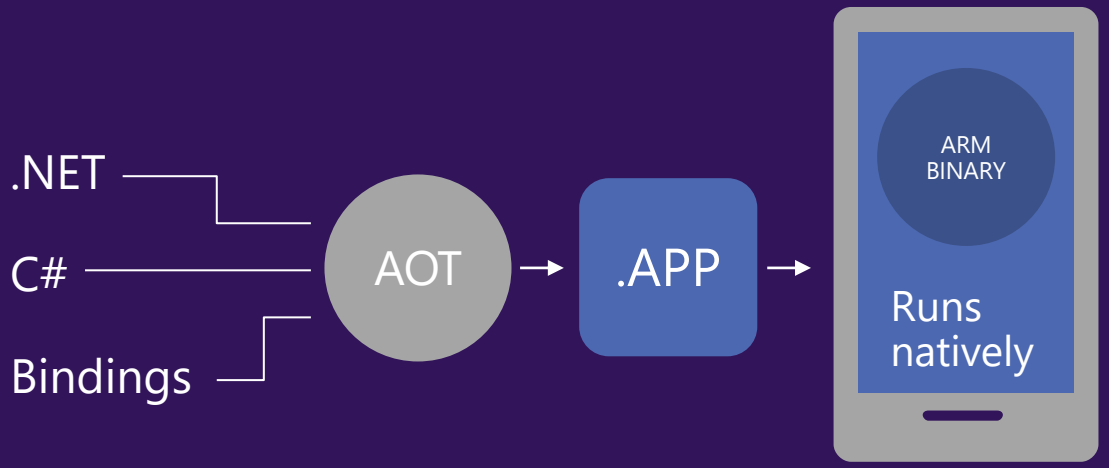
Are you a Microsoft 365 admin? Then you need the Microsoft 365 Admin app. Manager users, reset passwords, assign licenses, control notifications for service alerts, and connect with Microsoft Support.

Xamarin App Architecture

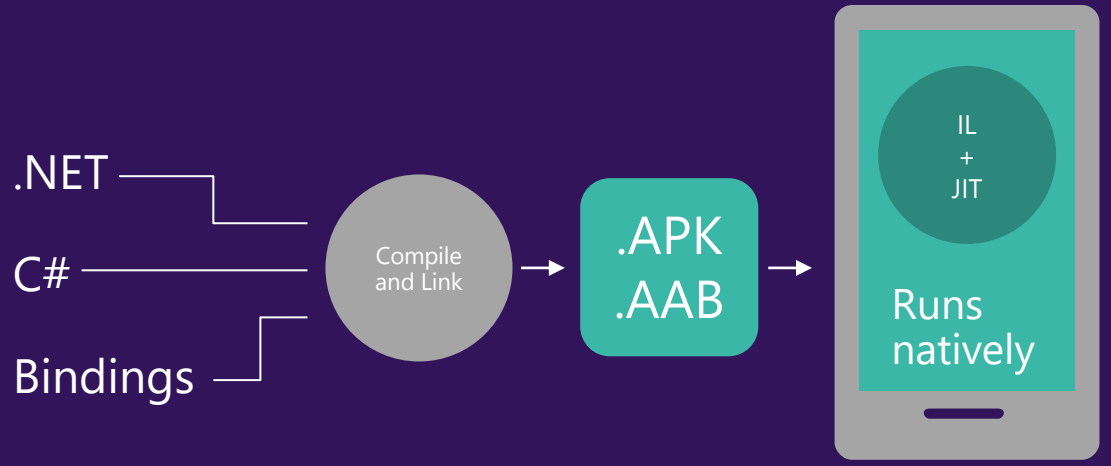


Shared C# codebase • 100% native API access • High performance

Native execution, native performance

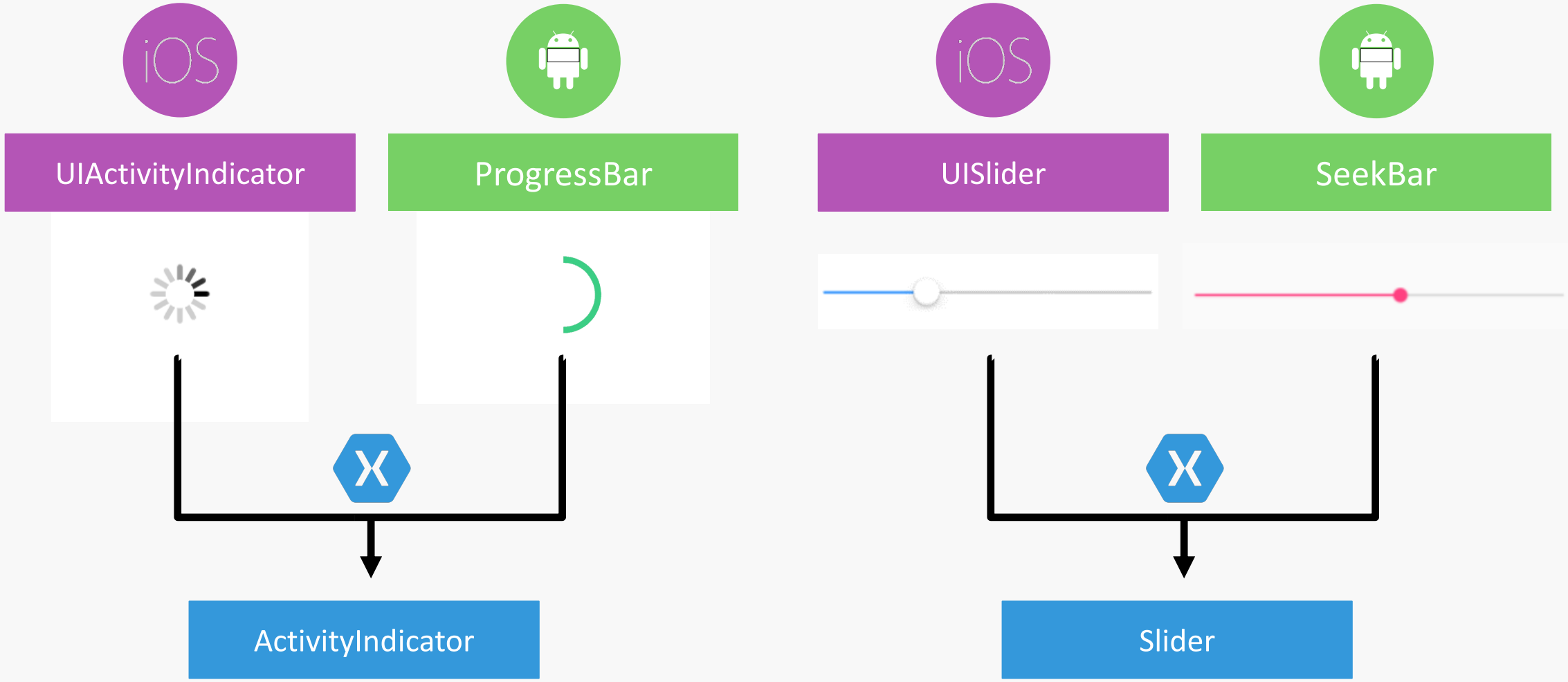


Xamarin.iOS does full Ahead Of Time (AOT) compilation to produce an ARM binary for Apple's App Store.



Xamarin.Android takes advantage of Just In Time (JIT) compilation on the Android device.

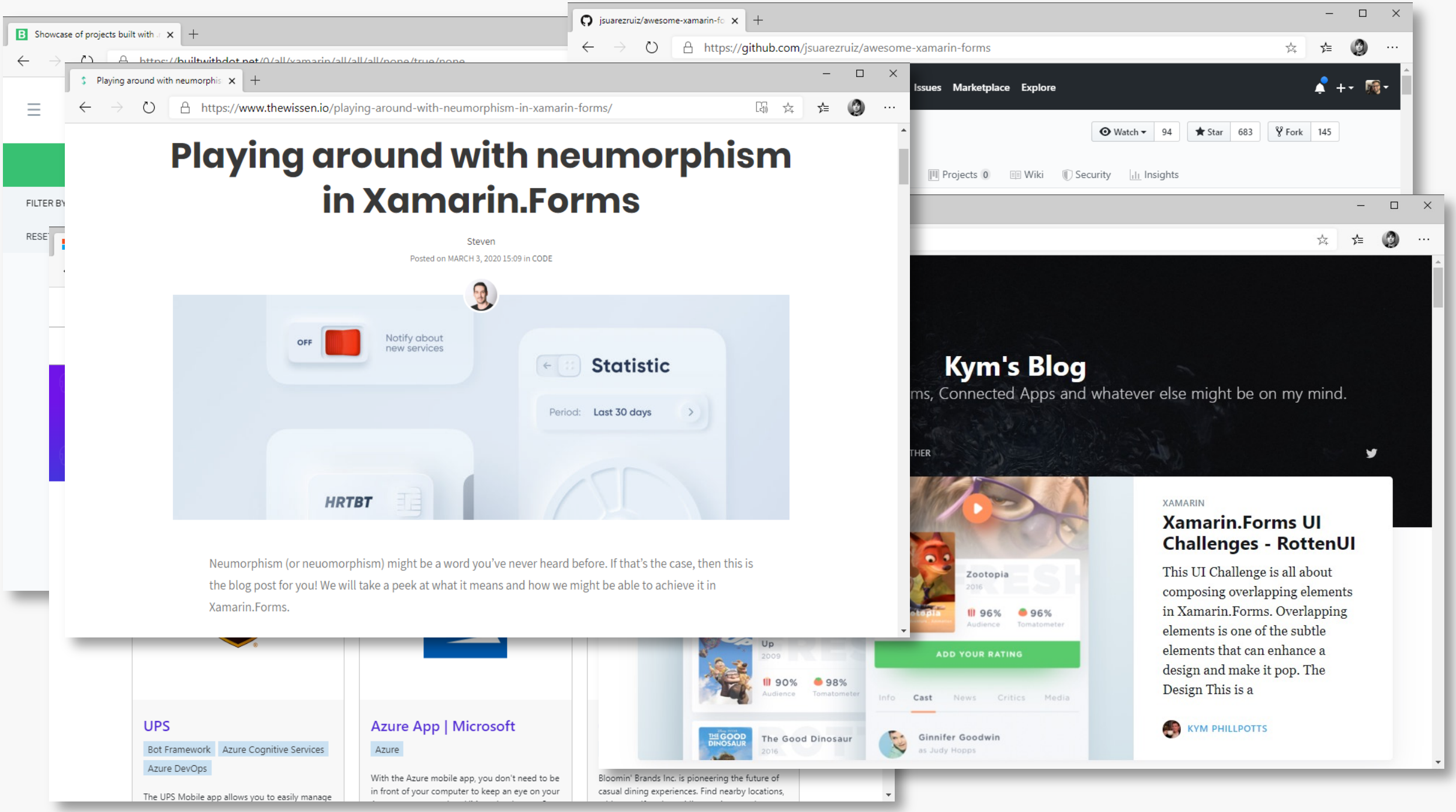
Native UI, even when shared

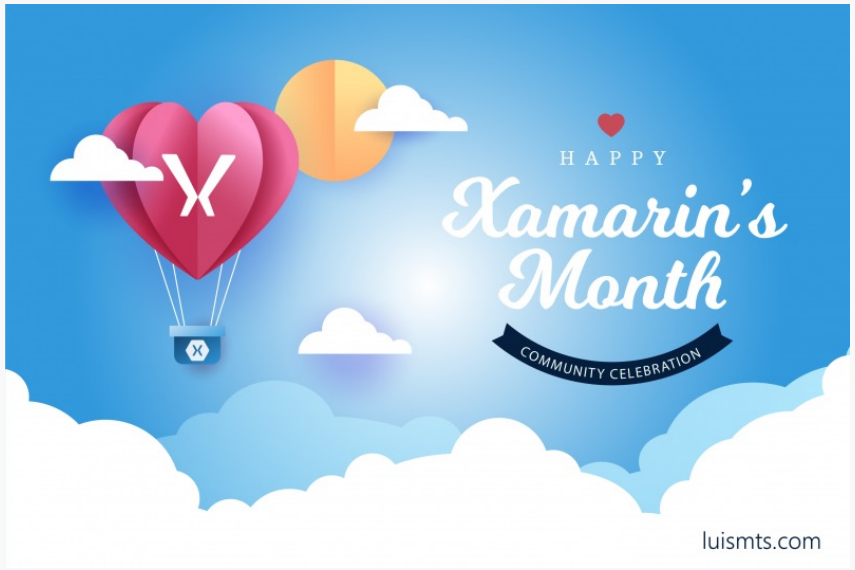


Xamarin in Action

.NET

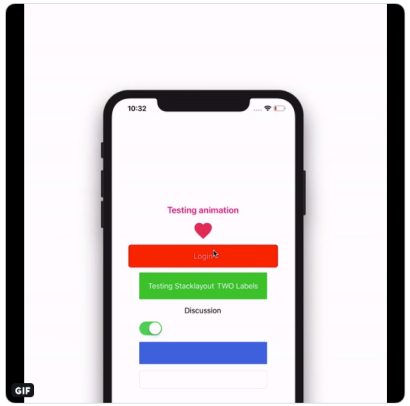



Showcase of projects built with
https://builtwithdot.net/0/all/xamarin/all/all/all/none/true/none
FILTER BY
RESET
UPS
Bot Framework
Azure Cognitive Services
Azure DevOps
The UPS Mobile app allows you to easily manage
Azure App | Microsoft
Azure
With the Azure mobile app, you don't need to be in front of your computer to keep an eye on your
Bloomin' Brands Inc. is pioneering the future of casual dining experiences. Find nearby locations,
jsuarezruiz/awesome-xamarin-fo
https://github.com/jsuarezruiz/awesome-xamarin-forms
Issues
Marketplace
Explore
Watch
94
Star
683
Fork
145
Projects 0
Wiki
Security
Insights
Playing around with neumorphis
https://www.thewissen.io/playing-around-with-neumorphism-in-xamarin-forms/
Playing around with neumorphism in Xamarin.Forms
Steven
Posted on MARCH 3, 2020 15:09 in CODE
OFF
Notify about new services
Statistic
Period:
Last 30 days
HRTBT
Neumorphism (or neuomorphism) might be a word you've never heard before. If that's the case, then this is the blog post for you! We will take a peek at what it means and how we might be able to achieve it in Xamarin.Forms.
Kym's Blog
ms, Connected Apps and whatever else might be on my mind.
THER
Zootopia
2016
96%
Audience
96%
Tomatometer
ADD YOUR RATING
Up
2009
90%
Audience
98%
Tomatometer
Info
Cast
News
Critics
Media
The Good Dinosaur
2016
Ginnifer Goodwin
as Judy Hopps
XAMARIN
Xamarin.Forms UI Challenges - RottenUI
This UI Challenge is all about composing overlapping elements in Xamarin.Forms. Overlapping elements is one of the subtle elements that can enhance a design and make it pop. The Design This is a
KYM PHILLPOTTS



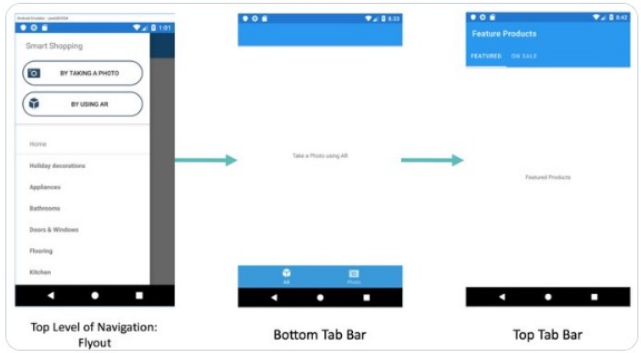
Charlin Agramonte @Chard003

I just wrote a new article about Plug and Play animations in Xamarin Forms xamgirl.com/plug-and-play-... #xamarinmonth #Xamarin @xamarinhq



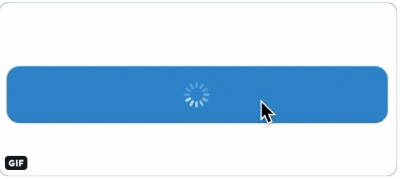
Alejandro Ruiz ➡ @XamarinSummit 2019 BR @alejandroruizva

Xamarin Month: Fall in Love with Xamarin.Forms Shell #Xamarin #xamarinmonth alejandroruizvarela.blogspot.com/2019/02/xamari...



John Taubensee @jtaubensee

Just wrote a short post on how to add an ActivityIndicator to a #XamarinForms Button. taubensee.net/activitybutton... #XamarinMonth



Daniel Monettelli L. @DanielMonetelli

Xamarin.Forms Update - Los enamorados CSS y BindableLayout - #DanielMonetelli #xamarinmonth #XamarinForms #CSS #BindableLayout #AdobeXD #UI #UX



Martin Zikmund #UnoConf @mzikmunddev

10 lovely things you should tell your friends about #Xamarin #xamarinmonth #day4 #XamarinForms #appdev #unoplatform #mvvmcross blog.mzikmund.com/2019/02/things...



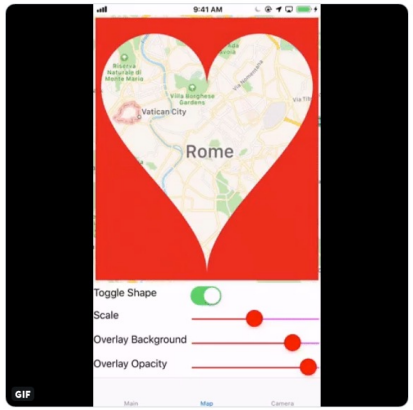
Marco Siccardi @msicc

Use NuGets for your common #Xamarin (Forms) code (and automate the creation process) #XamarinMonth bit.ly/2GciX04



Rendy Del Rosario @rdelrosario

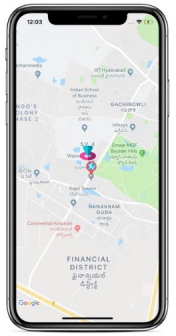
I just published a post about creating an Overlay view in Xamarin Forms xamboy.com/2019/02/23/ove... @xamarinhq #xamarinforms #xamarinmonth



Aritra Das @aritra_das

I just wrote a new article, titled "Using custom markers and Latitude Longitude bounds in Google Maps on Xamarin Forms" gameofthreads.online/custom-markers...

#XamarinForms #xamarinmonth #GoogleMaps #Android #iOS @xamarinhq @PlanetXamarin @davidortinau



pujolsluis @pujolsluis1

Hey Guys! just posted my article for the #xamarinmonth initiative about my recent experience helping out a member of our Xamarin Forms Dev Group on Facebook, she loved Xamarin and Prism @Elsa_AM4Y that now we got her first Xamarin Post for the community🥰🙌



Hussain Abbasi @HussainNAbbasi

New Blog Post: Stunning Animations In With Lottie.
My contribution to #XamarinMonth @ @xamarinhq #XamarinForms #Lottie



Luis Beltran @darkicebeam

Just published "#Azure Table Storage with #Xamarin Forms" on my blog as part of the #xamarinmonth initiative:

luisbeltran.mx/2019/02/02/azu...

#MVPBuzz



Luis Matos @luismatosluna

The future of #XamarinForms, an opinion of what may come in the following updates. Join me to see the Visual improvements in the development UI that #xamarin will have for us.

Post: bit.ly/2IzGOIV #XamarinMonth #XamarinUniverse #VisualControls @xamarinhq



Matthew Soucoup @CodeMillMatt

I wrote a special blog as part of #xamarinmonth!

5 Azure Offerings I ❤️For Xamarin Development!

codemilltech.com/5-azure-offeri...

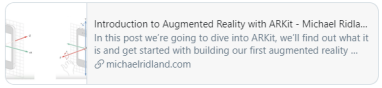
Kirk Patrick Junsay @KPjunsay

Profiling memory leaks in Xamarin Forms :) codesandchips.blog/2019/02/22/pro... #xamarinmonth

Michael Ridland @rid00z

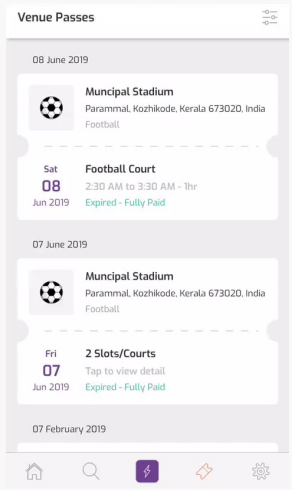
Replying to @luismatosluna

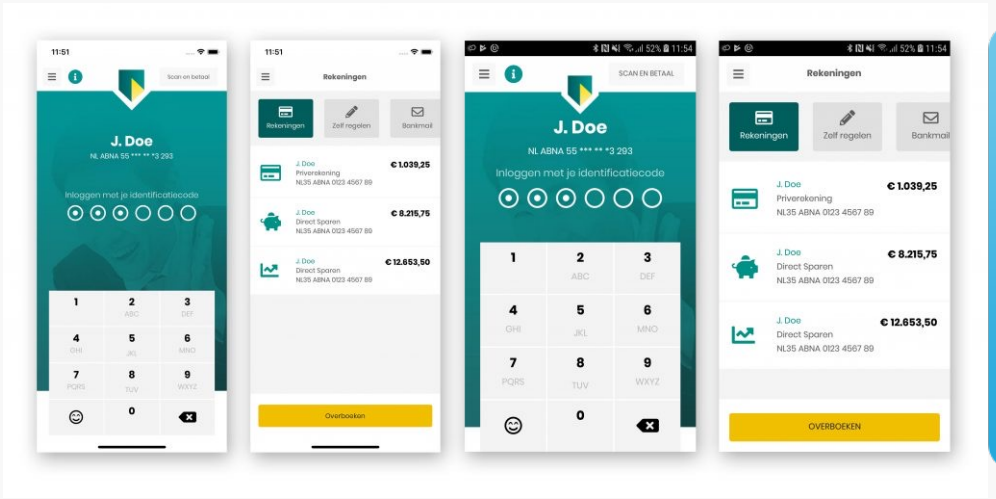
Yep, michaelridland.com/mobile/introdu... #xamarinmonth Introduction to Augmented Reality with ARKit



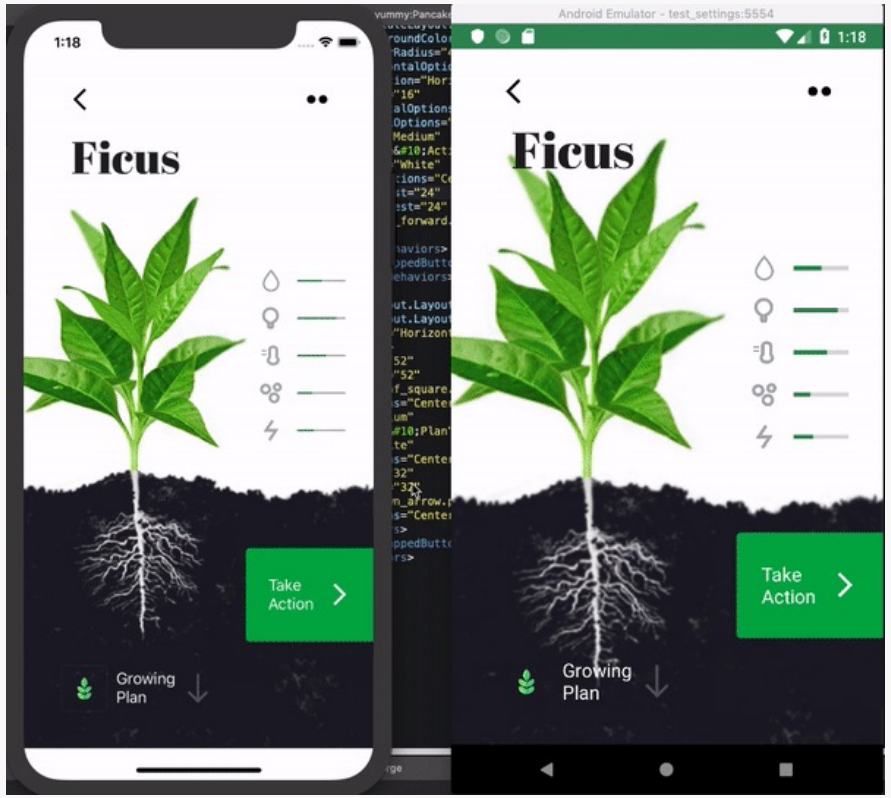
Sebastian | tsjdev-apps.de @tsjdevapps

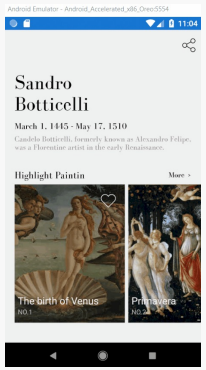
I just published Use FontAwesome in a app link.medium.com/1wyqcZyTjU #xamarin #xamarinmonth

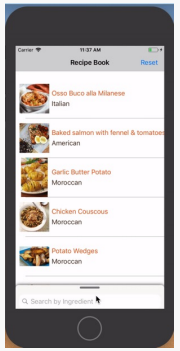

Venue Passes
08 June 2019
Muncipal Stadium
Parammal, Kozhikode, Kerala 673020, India
Football
Sat
08
Jun 2019
Football Court
2:30 AM to 3:30 AM - 1hr
Expired - Fully Paid
07 June 2019
Muncipal Stadium
Parammal, Kozhikode, Kerala 673020, India
Football
Fri
07
Jun 2019
2 Slots/Courts
Tap to view detail
Expired - Fully Paid
07 February 2019

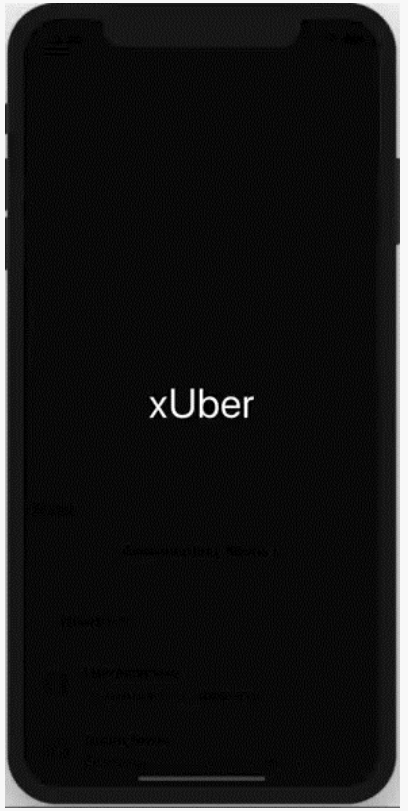

J. Doe
Inloggen met je identificatiecode
Rekeningen
Overboeken
SCAN EN BETAAL
OVERBOEKEN
€ 1.039,25
€ 8.215,75
€ 12.653,50

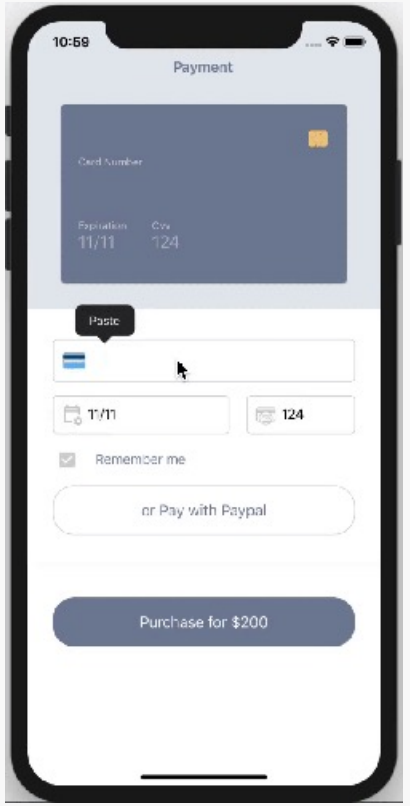

FEATURED BLOG
#XAMARINUIJULY

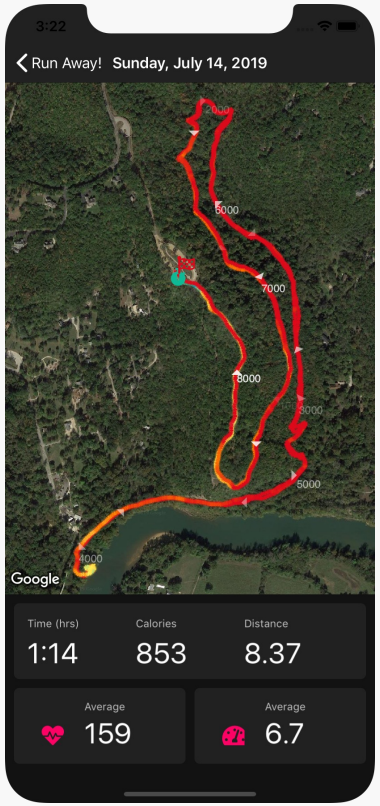

Ficus
Take Action
Growing Plan

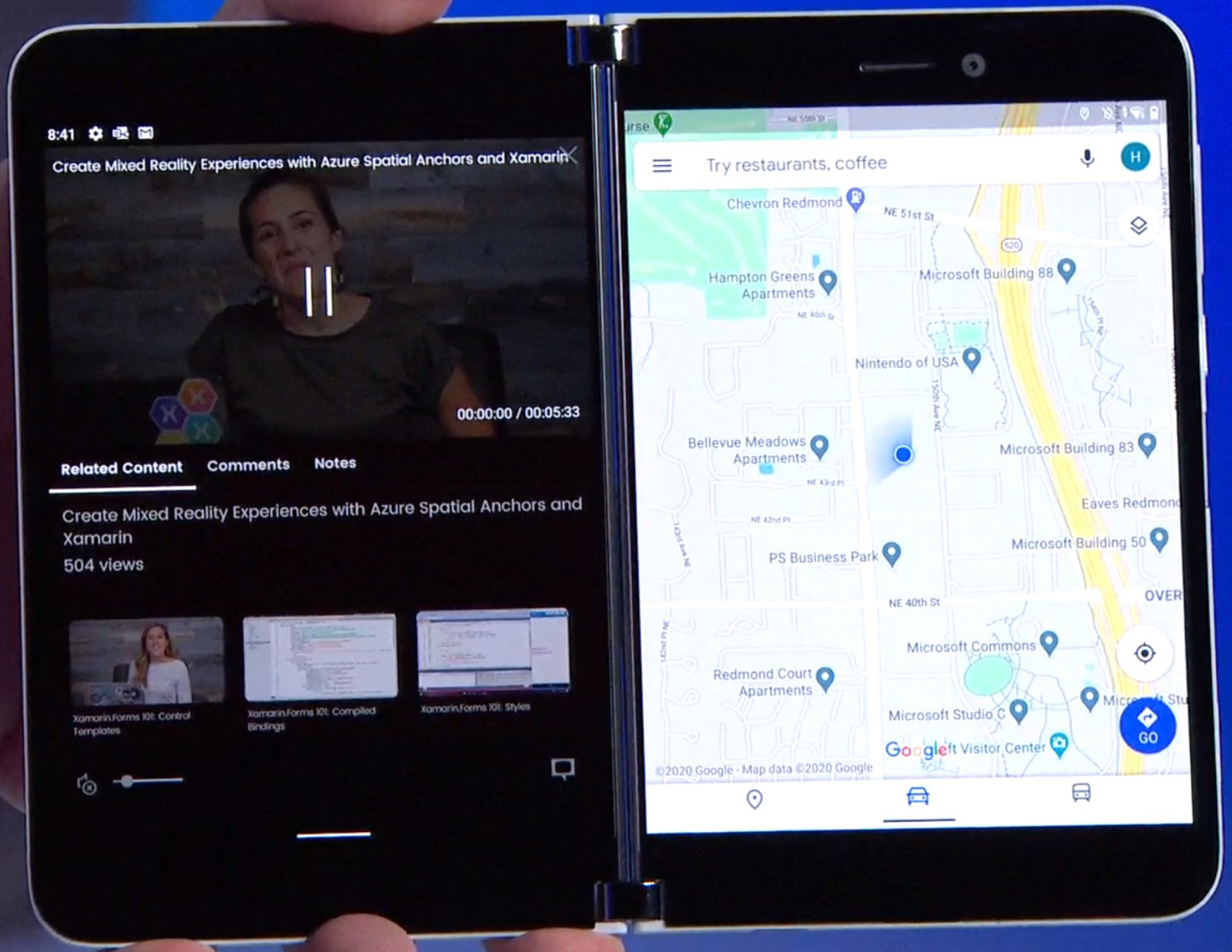

Sandro Botticelli
March 1, 1445 - May 17, 1510
Highlight Paintin
The birth of Venus
Primavera


Recipe Book
Reset


xUber


Payment
Paste
Remember me
or Pay with Paypal
Purchase for $200


Run Away! Sunday, July 14, 2019
Google
Time (hrs)
1:14
Calories
853
Distance
8.37
Average
159
Average
6.7


8:41
Create Mixed Reality Experiences with Azure Spatial Anchors and Xamarin
00:00:00 / 00:05:33
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NE 51st St
Hampton Greens Apartments
Microsoft Building 88
Nintendo of USA
Bellevue Meadows Apartments
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Eaves Redmond
PS Business Park
Microsoft Building 50
NE 40th St
Microsoft Commons
Redmond Court Apartments
Microsoft Studio C
Visitor Center
GO
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Surface Duo (Android) & Surface Neo (Win10X)





https://developer.microsoft.com/microsoft-365/virtual-events

https://aka.ms/xamarinconf .NET



Microsoft

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